RACE
Regular Analysis Through Continuous Exercise

The p-Block Elements : RACE-01

Group 13 Elements; The Boron Family (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

1. Orthoboric acid –

(A) Donate proton to form $H_2BO_3^{\ominus}$ (B) Accept proton of form $H_4BO_3^{\oplus}$

(C) Donate $OH^{\ominus}$ to form $H_2BO_2^{\oplus}$ (D) Accept OH^- to form $[B(OH)_4]^{\ominus}$

CPB121

2. Alumina is insoluble in water because:

(A) It is a covalent compound.

(B) It has high lattice energy and low heat of hydration

(C) It has low lattice energy and high heat of hydration

(D) $Al^{3\oplus}$ and $O^{2\ominus}$ ions are not excessively hydrated.

CPB122

Preparation of Boron Some Important compounds of Group-13 and their physical and chemical properties (Borax, Diborane, Orthoboric Acid, Alums)

3. $Na_2B_4O_7.10H_2O \xrightarrow{\Delta} P \xrightarrow{\Delta} Q + R$

P, Q and R in the reaction are :

(A) $P = NaBO_2$, $Q = B_2O_3$, $R = B(OH)_3$ (B) $P = B_2O_3$, $Q = NaBO_2$, $R = B(OH)_3$

(C) $P = Na_2B_4O_7$, $Q = NaBO_2$, $R = B_2O_3$ (D) $P = Na_2B_4O_7$, $Q = B_2O_3$, $R = H_3BO_3$

CPB134

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Uses of Boron and Aluminium and their compounds

4. Borax is used as cleansing agent because on dissolving in water it gives –

(A) Alkaline solution (B) Acidic solution

(C) Belching solution (D) Colloidal solution

CPB119

Group 14 Elements; The Carbon Family (Occurance , Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

5. Which element is metallic in nature:

(A) Si (B) Sn (C) Ge (D) C

CPB141

Allotropes of Carbon

6. Select the incorrect statement:

(A) Hybridisation of C in diamond and graphite is sp^3 and sp^2 respectively.

(B) Fullerenes are artificial allotrope of carbon.

(C) The Van Der Waals force (weak force) is present between two adjacent layers in graphite

(D) Diamond is bad conductor of heat

CPB146

For More Join: @IITJEE_Advanced

Some important Compounds of Carbon: CO and CO_2

7. Which of the following statement is false:

(A) Dry ice is solid CO_2 (B) CO_2 is weakly acidic

(C) CO_2 and SiO_2 are linear molecules (D) CO_2 is a gas while SiO_2 is solid

CPB148

8. Select the correct statements:

(A) CO_2 is a colourless, odourless and heavy gas.

(B) It is slightly soluble in water under ordinary pressure but highly soluble at high pressure

(C) Estimation of CO is done by I_2O_5

(D) CO is absorbed in ammoniacal CuCl solution.

CPB150

Silicon and its Compounds SiO_2

9. In silicon dioxide:

(A) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms

(B) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms

(C) Silicon atom is bonded to two oxygen atoms

(D) There are double bonds between silicon and oxygen atoms

CPB155

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Silicones

10. Silicones have the general formula.

(A) $SiO_4^{4\ominus}$ (B) $Si_2O_7^{6\ominus}$ (C) $(R_2SiO)_n$ (D) $\left(SiO_3\right)_n^{2\ominus}$

CPB153

11. Glass or Silica is soluble in –

(A) $HClO_4$ (B) HF (C) Aqua-regia (D) H_2SO_4

CPB154

12. Among the following substituted silanes, the one which will give rise to cross linked silicone polymer on hydrolysis is

(A) R_4Si (B) $RSiCl_3$ (C) R_2SiCl_2 (D) R_3SiCl

CPB156

Silicates

13. $\left(Si_2O_5\right)_n^{2n\ominus}$ anion is obtained when.

(A) no oxygen of a $SiO_4^{4\ominus}$ tetrahedron is shared with another $SiO_4^{4\ominus}$ tetrahedron

(B) one oxygen of a $SiO_4^{4\ominus}$ tetrahedron is shared with another $SiO_4^{4\ominus}$ tetrahedron

(C) two oxygen of a $SiO_4^{4\ominus}$ tetrahedron is shared with another $SiO_4^{4\ominus}$ tetrahedron

(D) three oxygen of a $SiO_4^{4\ominus}$ tetrahedron is shared with another $SiO_4^{4\ominus}$ tetrahedron

CPB157

For More Join: @IITJEE_Advanced

14. The silicate anion in the mineral is a chain of three $SiO_4^{4\ominus}$ tetrahedral, that share corners with adjacent tetrahedral. The charge of the silicate anion is

(A) –4 (B) –8 (C) –6 (D) –2

CPB158

Comprehension : (Q.15 to Q.17)

Column-I (Unit formula of different silicates)	Column-II (Number of unshared oxygen in per tetrahedral unit)	Column-III (Number of shared oxygen in per tetrahedral unit)
(I) $(Si_4O_{11}^{6\ominus})_n$	(i) 2	(P) 2
(II) $(Si_2O_5^{2\ominus})_n$	(ii) 1	(Q) 3
(III) $(SiO_2)_n$	(iii) 0	(R) 4
(IV) $(SiO_3^{2\ominus})_n$	(iv) 3	(S) 1

15. For the sheet silicate the only CORRECT combination is:-

(A) (II), (i), (P) (B) (II), (ii), (Q) (C) (IV), (ii), (R) (D) (III), (iii), (P)

CPB159

16. For 3D silicate (silica) the only CORRECT combination is :-

(A) (II), (ii), (Q) (B) (III), (iv), (R) (C) (III), (iii), (R) (D) (III), (iv), (S)

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CPB160

17. For single chain silicate the only CORRECT combination is :-

(A) (I), (iii), (R) (B) (IV), (ii), (P) (C) (III), (ii), (Q) (D) (IV), (i), (P)

CPB161

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	A	B	D	C	ABCD	A	C
Question	11	12	13	14	15	16	17			
Answer	B	B	D	B	B	C	D			

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

The p-Block Elements : RACE-02

Group 15 Elements (Occurance, Electronic Configuration, Atomic Radii, Ionisation Enthalpy, Electronegativity, Catenation, Allotropy, Physical Properties)

1. Which element does not show Allotropy?

(A) N (B) P (C) As (D) Sb

CPE090

2. From Arsenic to Bismuth, only a small increase in covalent radius is observed because -

(A) Increase in Nuclear charge

(B) Increase in number of shells

(C) Poor shielding by d and f orbitals electrons

(D) Increase in number of electrons.

CPE091

3. Total number of metalloids in Pnicogens?

CPE110

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Dinitrogen Preparation and Properties

4. Which of the following statement is **INCORRECT** about N_2 ?

(A) It is a odourless & non-toxic gas

(B) It is an colourless and tasteless gas

(C) It is highly soluble in water

(D) It is inert at room temperature

CPE096

5. From the given reaction

$(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta} x + y + H_2O$

Sum of Bond order of 'x' and oxidation state of compound 'y'?

CPE109

6. From the given reaction.

Sodium azide $\xrightarrow{\Delta}$ x Na + yN_2

Multiplication of x and y in the above reaction.

CPE111

Ammonia Preparation and Properties

7. Ammonia gas can be dried by –

(A) Anhydrous $CaCl_2$ (B) Conc. H_2SO_4 (C) P_2O_5 (D) Quick lime

CPE098

For More Join: @IITJEE_Advanced

8. What are the necessary condition in accordance with Le-chatelier's principle for production of NH_3 by Haber's process ?

(A) High temperature and Low pressure

(B) High temperature and High pressure

(C) Low temperature and Low pressure

(D) Low temperature and High pressure

CPE099

9. Nitrogenous fertilizers are:

(A) Ammonium nitrate (B) Urea

(C) Ammonium phosphate (D) All of the above

CPE102

10. The products of hydrolysis of NCl_3 and PCl_3 are respectively

(A) NaOCl (B) H_3PO_3 (C) PH_3 (D) NH_3

CPE104

11. Find the value of $|y - x|$ if

x = Number of σ bonds in urea molecule

y = Total number of nonbonding e– pairs in urea molecule

CPE112

12. Total number of ammonia moles produce in hydrolysis reaction of two mole of Lithium Nitride?

CPE113

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Oxides of Nitrogen

13. What is INCORRECT about N_2O_5 ?

(A) It is anhydride of HNO_3

(B) It is a powerful oxidizing agent

(C) Solid N_2O_5 is called Nitronium nitrate.

(D) Structure of N_2O_5 contains zero coordinate Bond

CPE114

14. The Nitrogen oxide that does not contain N — N bond is-

(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5

CPE115

Nitric Acid Preparation and Properties

15. HNO_3 is prepared on large scale by -

(A) Haber's process (B) Ostwald's process

(C) Contact process (D) All of above

CPE116

16. 98% HNO_3 can be achieved by dehydration with:

(A) P_2O_5 (B) Conc. H_2SO_4 (C) Anhyd. $CaCl_2$ (D) Quick lime

CPE117

For More Join: @IITJEE_Advanced

17. Which of the following does not produce NO_2 gas with conc. HNO_3 ?

(A) Cu (B) I_2 (C) Ag (D) Au

CPE119

18. The catalyst used in Ostwald's process is -

(A) Ni (B) Fe (C) Pt (D) V_2O_5

CPE120

19. How many of the following will be oxidised by HNO_2 ?

$I^{\ominus}$, $Fe^{2\oplus}$, $Sn^{2\oplus}$, $SO_3^{2\ominus}$

CPE132

20. Number of oxides of nitrogen, which produe HNO_3 when dissolved in water.

N_2O, NO, NO_2, N_2O_4, N_2O_5

CPE134

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	2	C	6	6	D	D	D	BD
Question	11	12	13	14	15	16	17	18	19	20
Answer	3	2	D	D	B	B	D	C	4	3

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

The p-Block Elements : RACE-03

Allotropes of Phosphorus; Preparation and Properties

1. Which of the following is NOT an allotrope of phosphorus?

(A) White Phosphorus (B) Green Phosphorus
(C) Red Phosphorus (D) Black Phosphorus

CPE124

2. White phosphorus is less stable because of–

(A) Large size (B) Low boiling and melting point
(C) Angular strain (D) Electronic repulsion

CPE125

3. Which of the following is polymeric?

(A) Red P (B) White P
(C) α-black P (D) β-black P

CPE126

4.

Column – I	**Column – II**
(a) α-Black Phosphorus	(P) Prepared by heating white phosphorus at 473 K under high pressure
(b) β-Black Phosphorus	(Q) Does not glow in dark
(c) Red phosphorus	(R) Soluble in CS_2
(d) White phosphorus	(S) Prepared by heating red Phosphorus at 803 K in a sealed tube

(A) a → P; b → S; c → Q; d → R (B) a → S; b → P; c → Q; d → R
(C) a → S; b → P; c → R; d → Q (D) a → R; b → S; c → P; d → Q

CPE127

5. Correct statements about white Phosphorus

(A) It is soft waxy solid
(B) It is white when pure but attains yellow colour on standing.
(C) It is insoluble in water
(D) It is soluble in CS_2

CPE130

6. Total number of P–P linkage in P_4 molecule.

CPE136

Phosphine Preparation and Properties

7. Which of the following are correct for PH_3

(A) Slightly soluble in water (B) Soluble in CS_2
(C) Soluble in organic solvent (D) Highly soluble in water

CPE150

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For More Join: @IITJEE_Advanced

PCl_3 Preparation and Properties

8. PCl_3 can be obtained by reaction of :
(A) $P_4 + SOCl_2$ (B) $P_4 + SO_2Cl_2$ (C) Both (D) None

CPE141

P_4O_6 Preparation and Properties

9. Which of the following statements are correct for P_4O_6 :
(A) No. of P – O bonds are 6 (B) No. of P – O – P bonds are 6
(C) No of P – O bonds are 12 (D) No of P – O – P bonds are 12

CPE152

P_4O_{10} Preparation and Properties

10. P_4O_{10} is used extensively as a :
(A) Dehydrating agent (B) Catalytic agent
(C) Reducing agent (D) Preservative

CPE146

Simple and Mixed Oxides

11. The correct order of acidic strength is:
(A) $Cl_2O_7 > SO_3 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$
(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

CPE173

Allotropes of Sulphur

12. The stable form of sulphur at room temperature is :
(A) Monoclinic sulphur (B) Rhombic sulphur
(C) Cyclo sulphur (D) Sulphur vapours

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CPE188

Sulphuric Acid(H_2SO_4): Preparation and Properties

13. H_2SO_4 has very high corrosive action on skin because :
(A) It reacts with proteins
(B) It acts as an oxidising agent
(C) It acts as a dehydrating agent
(D) It acts as dehydrating agent and absorption of water is highly exothermic

CPE201

14. The optimum conditions for maximum yield from contact process are :
(A) Low temperature and low pressure
(B) High temperature and low pressure
(C) Low temperature and high pressure
(D) High temperature and high pressure

CPE202

15. $S + H_2SO_4$ (conc.) $\rightarrow$ Gas A + $2H_2O$
$C + H_2SO_4$ (conc.) $\rightarrow$ Gas A + $CO_2 + 2H_2O$
Here gas A will be :
(A) SO_2 (B) SO_3 (C) S_2 (D) Both A and B

CPE203

For More Join: @IITJEE_Advanced

16. During contact process, SO_3 is NOT absorbed directly in water because :
(A) Reaction will be highly exothermic
(B) Acid fog is formed, which is difficult to condense
(C) SO_3 decomposes to SO_2 and O_2
(D) Solubility of SO_3 is very less in water.

CPE204

Sodium Thiosulphate and its applications

17. When an inorganic compound reacts with SO_2 in aqueous medium, produces (A). (A) on reaction with Na_2CO_3, gives compound (B) which with sulphur, gives a substance (C) used in photography. Compound (C) is :
(A) Na_2S (B) $Na_2S_2O_7$ (C) Na_2SO_4 (D) $Na_2S_2O_3$

CPE205

18. In Thiosulphuric acid :
(A) There is only one replaceable hydrogen atom
(B) One S atom is in +2 and other sulphur atom is in +4 oxidation state
(C) There is a S = S linkage present
(D) Each sulphur atom is in identical oxidation state

CPE207

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	A	B	ABCD	6	ABC	A	BC	A
Question	11	12	13	14	15	16	17	18		
Answer	A	B	D	C	A	B	D	C		

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

The p-Block Elements : RACE-04

Anomalous behaviour of Fluorine; Chemical properties of Group 17 Elements (Reactivity towards Hydrogen, Oxygen, Metals and other Halogens)

1. Total number of unpaired electrons in second excited state of Bromine?

CPE236

Chlorine: Preparations and Properties

2. When I_2 is passed through KCl, KF, KBr sulutions :

(A) Cl_2 and Br_2 are evolved
(B) Cl_2 is evolved
(C) Cl_2, Br_2, F_2 are evolved
(D) No gas is evolved

CPE221

3. Which of following will not displace halogen from the solution of the halide?

(A) Br_2 added to NaI
(B) Br_2 added to NaCl
(C) F_2 added to KCl
(D) Cl_2 added to NaF

CPE229

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Chlorine: Preparations and Properties

4. From given reaction :

$$4HCl(g) + O_2(g) \xrightarrow[723k]{CuCl_2} X\,Cl_2(g) + y\,H_2O$$

Correct statement about :

(A) x = 2
(B) y = 1
(C) This is reaction for preparation of Cl_2 by Deacon's process.
(D) None of these

CPE231

5. Correct statement about Cl_2 :

(A) It is a greenish yellow gas.
(B) It is 2.5 times heavier than air.
(C) It is not soluble in water.
(D) Having pleasant smell.

CPE232

6. $\underset{\text{(Excess)}}{8NH_3} + 3Cl_2 \longrightarrow x + 6\,NH_4Cl$

$NH_3 + \underset{\text{(Excess)}}{3Cl_2} \longrightarrow y + 3\,HCl$

Then choose the correct statement(s) is/are :

(A) x = NCl_3 (B) y = N_2 (C) x = N_2 (D) y = NCl_3

CPE233

For More Join: @IITJEE_Advanced

Hydrochloric Acid(HCl): Preparation and Properties

7. Aqua regia is:

(A) 3 : 2 mixture of HCl and HNO_3 (B) 3 : 1 mixture of HCl and HNO_3

(C) 3 : 1 mixture of H_2SO_4 and HCl (D) 1 : 1 mixture of HCl and HNO_3

CPE241

8. $2KMnO_4 + 16HCl \longrightarrow X + KCl + 8H_2O + 5Cl_2$ The oxidation state of Mn in 'X' is:

(A) +1 (B) +2 (C) +4 (D) +7

CPE243

9. Which of the following are incorrect?

(A) Blue litmus turns into Red litmus in the presence of perfectly dry HCl.

(B) Aqua regia is a mixture of conc. HCl and conc. HNO_3 in ratio of 1 : 3.

(C) HCl is dried over P_4O_{10}.

(D) HCl can't be dried over CaO because HCl reacts with CaO chemically.

CPE257

Oxoacides of Halogens: Preparation and Properties

10. Which of the following are correct?

(A) $HOClO_2$ is more acidic than $HClO_2$.

(B) Oxidation state of oxygen is –2 in $HClO_4$.

(C) Lone pair of 'Cl' is present in sp^2 hybridization orbital in $HClO_3$.

(D) $HClO_4$ is most acidic oxyacid among all the oxyacids of Chlorine.

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CPE258

Interhalogen Compounds

11. $Cl_2 + F_{2(Excess)} \xrightarrow{573K}$ (A)

Shape of the compound (A) is:

(A) Linear (B) Tetrahedral

(C) Trigonal bipyramidal (D) T-shaped

CPE247

12. Which of following are correct?

(A) BrF is a pale Brown solid.

(B) ICl_3 is an orange colour solid.

(C) All the interhalogens containing fluorine are colourless compounds.

(D) IBr is thermally more stable than BrCl.

CPE259

Poly Halides; Pseudo Halides halogen Species

13. Which of the following is an inter – pseudo – halogen (Pseudo halogen analogues of inter – halogen)?

(A) C_2N_2 (B) BrF_2 (C) ICN (D) HSCN

CPE251

For More Join: @IITJEE_Advanced

Chemical properties of Group 18 Elements

14. Correct statement(s) regarding physical properties of noble gases :

(A) All the noble gases are monoatomic.

(B) They are colourless, odourless tasteless.

(C) They are highly soluble in water.

(D) Helium has lowest boiling point (4.2K) of any known substance.

CPE283

Compounds of Xenon with Oxygen and Fluorine, Chemical Properties

15. XeF_2 reacts with SbF_5 to form :

(A) $[XeF]^{\oplus}\,[SbF_6]^{\ominus}$ (B) $[XeF_3]^{\ominus}\,[SbF_4]^{\oplus}$ (C) $Xe^{\oplus}\,[PtF_6]^{\ominus}$ (D) XeF_4

CPE271

16. Identify the number of reaction which give Xe as one of the product?

(A) (i) $XeF_4 + SF_4 \longrightarrow$ (B) (ii) $XeF_6 + 3H_2O \longrightarrow$

(C) (iii) $XeF_4 + H_2O \longrightarrow$ (D) (iv) $XeF_6 + H_2O \xrightarrow{\text{excess } OH^{\ominus}}$

CPE284

Group 18: Uses

17. Helium is added to Oxygen used by deep sea divers because :

(A) It is less soluble in blood than nitrogen under high pressure

(B) It is lighter than nitrogen

(C) It is readily miscible with oxygen

(D) It is less poisonous than nitrogen

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CPE276

18. Select the correct statements(s).

(A) Ne bulbs are used in botanical gardens and green houses.

(B) Ar is use to provides inert atmosphere in high temperature weldings.

(C) Xenon and krypton used in light bulbs designed for special purposes.

(D) None of these

CPE287

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	5	D	BD	AC	AB	CD	B	B	ABC	ABD
Question	11	12	13	14	15	16	17	18		
Answer	D	BCD	C	ABD	A	ACD	A	ABC		

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